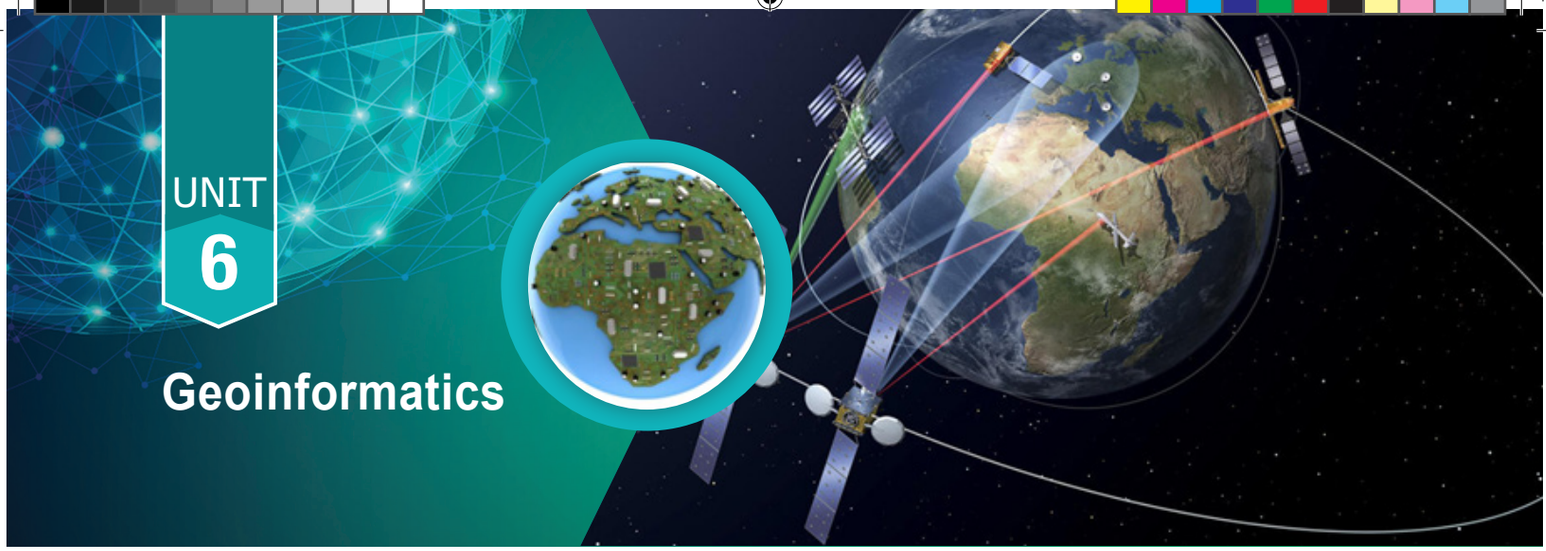


UNIT 6

Geoinformatics

Unit Overview

6.1 Introduction

6.2 Remote sensing

6.3 Geographic Information System (GIS)

6.4 Global Navigation Satellite System (GNSS)

Learning Objectives

- *To understand major divisions of Geoinformatics- Remote sensing GNSS, GIS*
- *To value the application of remote sensing in various fields*
- *To learn to define GIS and understand its usage, in general, learn to calculate spatial data and its analysis (Raster and vector data).*

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6.1 Introduction

Geoinformatics is the integration of remote sensing, Global Navigation Satellite System and Geographic Information System dealing with spatial information. The advent of remote sensing, Global Navigation Satellite System and Geographic Information System has made significant changes in surveying and map making. A basic understanding of these components is crucial for carrying out various types of surveys, navigation, hydrology, disaster management, etc.

The field of Remote Sensing and GIS has become exciting with rapidly expanding opportunities. Geoinformatics has three major components namely remote sensing, GIS and GNSS.

6.2 Remote sensing

Remote sensing is an integrated discipline encompassing some branches of arts, science and technology of collecting information about the terrestrial objects using camera and sensor system.

Elements of Remote Sensing

1. Energy Source

The primary requirement for remote sensing is to have an energy service, which provides electromagnetic energy to the target of interest. The sun being a major source of energy, radiation and illumination having a sharp power allows capturing reflected light with conventional cameras and films.

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